

Feedback Control Systems (EE3004)

Sessional-I Exam

Date: February 21st 2026

Total Time (Hrs): 1

Course Instructor

Total Marks: 30

Dr. Omer Saleem & Ms. Tamania Javed

Total Questions: 3

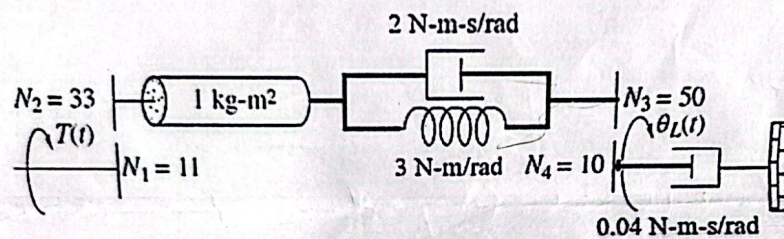
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Student Signature

Attempt all the questions.

CLO # 1: Develop mathematical models of physical systems using transfer functions and state-space methods.

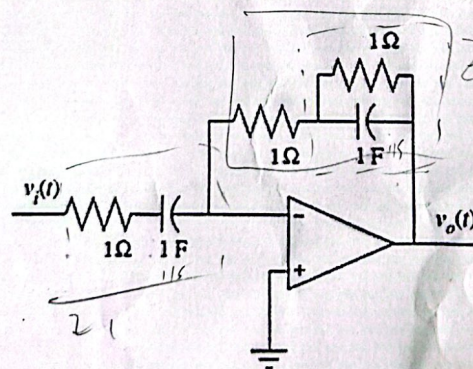
Q1: Formulate the transfer function, $G(s) = \theta_L(s) / T(s)$, for the geared mechanical system shown in the Figure below. [10 marks]



$$\frac{r_2}{r_1} = \frac{N_2}{N_1}$$

CLO # 1: Develop mathematical models of physical systems using transfer functions and state-space methods.

Q2: Formulate the transfer function $G(s) = V_o(s) / V_i(s)$ of the following inverting amplifier circuit. Hence, use this transfer function to evaluate $v_o(t)$, if a unit-step signal is applied at $v_i(t)$. [10 marks]

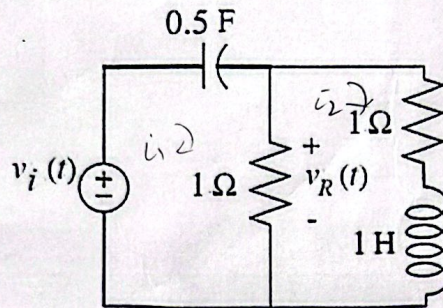


$$1/5$$

$$\frac{-27}{21}$$

CLO # 1: Analyze the transient and steady state response of a feedback system

Q3: Assume that currents $i_1(t)$ and $i_2(t)$ are flowing in loop 1 and loop 2 of the following circuit, respectively. The input of the circuit is $V_i(t)$. The voltage across the capacitor is $V_c(t)$. The output of the circuit is $V_R(t)$. Develop the state-space model of the system, identifying the state variables as well as the matrices A, B, C, and D. [10 marks]



$$y(t) = v_R(t)$$

$$u(t) = \begin{bmatrix} v_c \\ \dot{v}_c \end{bmatrix}$$

$$\dot{v}_c = \dot{v}_2$$

$$v_i = v_c + v_R$$